ALLIANCE UNIVERSITY

NAAC GRADE A+ ACCREDITED UNIVERSITY

ALLIANCE SCHOOL OF COMPUTER APPLICATIONS

Bachelor of Computer Applications (BCA) — IV Semester

PROJECT WORK

E1CAA311 - Data Warehousing and Mining

Sales Performance Analysis using Power BI

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COURSE

E1CAA311 - Data Warehousing and Mining

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Abstract

This project focuses on designing and implementing a data warehouse using Power BI. The main objective is to analyze sales data and generate meaningful insights through interactive dashboards. The dataset is collected from Kaggle and processed using Power Query for cleaning and transformation. Various visualizations such as bar charts, pie charts, line charts, and maps are used. The project helps in understanding business performance and supports better decision-making.

Introduction

Data visualization is an important process for understanding large datasets. Organizations use tools like Power BI to convert raw data into meaningful insights. This project aims to build dashboards to analyze sales data and identify trends and patterns.

Problem Statement

The problem is to analyze sales data and identify key business insights such as top-performing categories, customer segments, and geographical distribution. The goal is to transform raw data into useful information for business decisionmaking.

Dataset Description

The dataset used is the Superstore Sales dataset obtained from Kaggle.

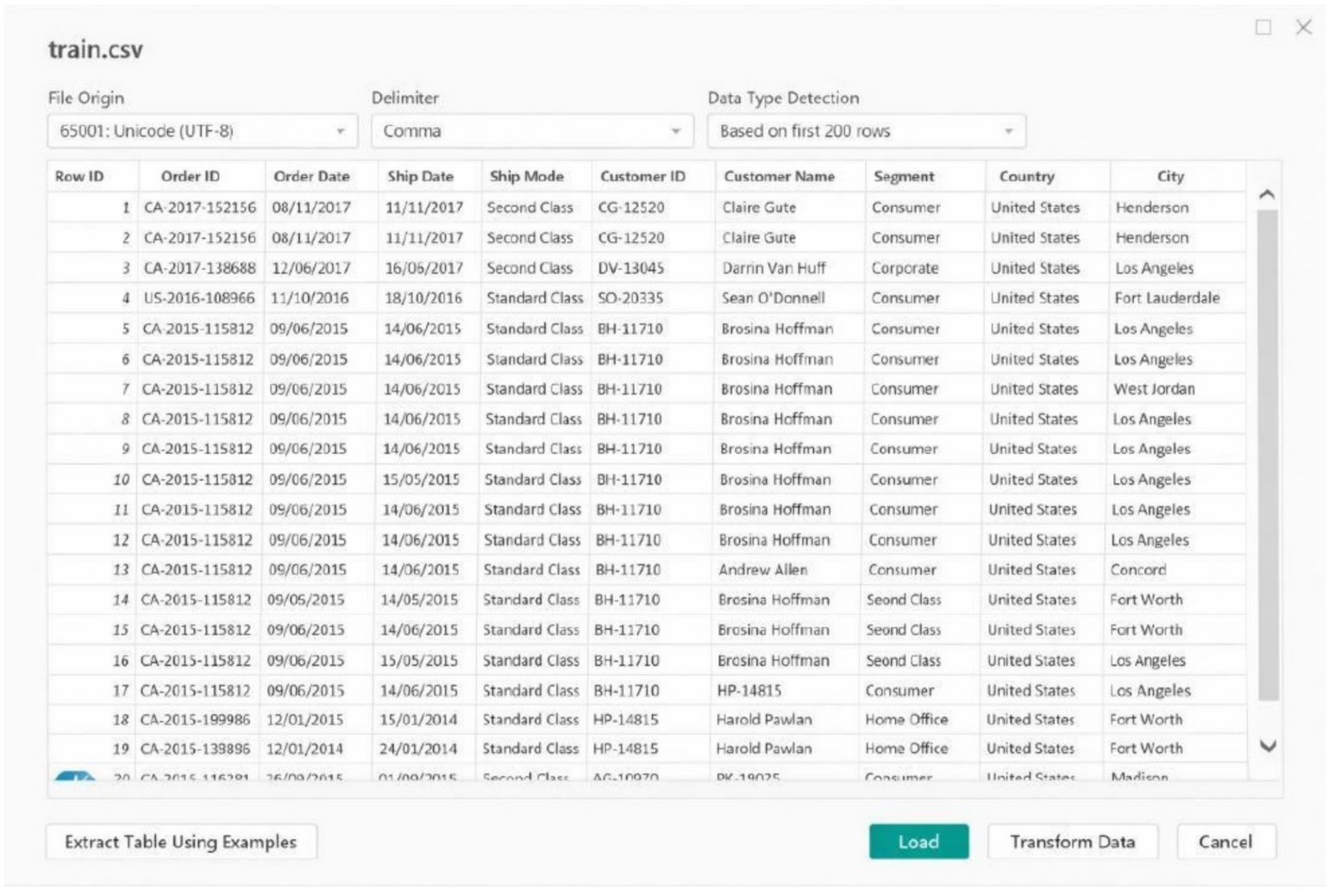
Number of Rows: Approximately 9000+ Number
of Columns: Around 10–15

Features include:
Order ID
Customer Name
City, State
Category, Segment
Sales
Order Date

Data Preprocessing (Power Query)

The following steps were performed:

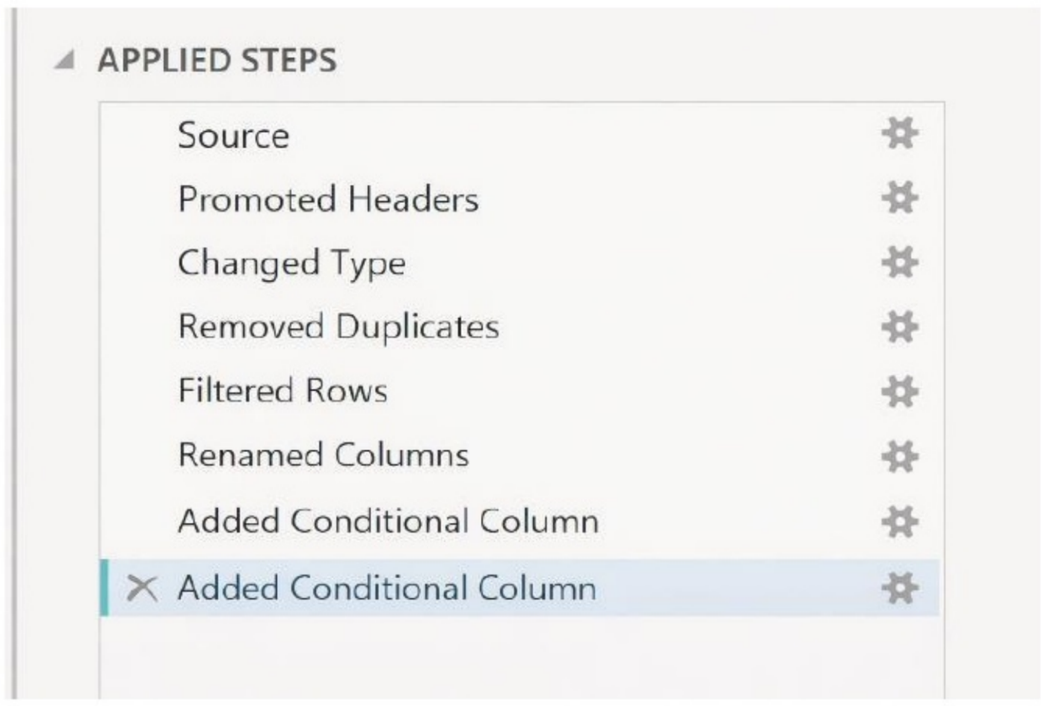
Removed duplicate records
Handled missing values
Renamed columns
Changed data types
Filtered data
Added conditional columns
Removed unnecessary columns



Row ID	Order ID	Order Date	Ship Date	Ship Mode	Customer ID	Customer Name	Segment	Country	City
1	CA-2017-152156	08/11/2017	11/11/2017	Second Class	CG-12520	Claire Gute	Consumer	United States	Henderson
2	CA-2017-152156	08/11/2017	11/11/2017	Second Class	CG-12520	Claire Gute	Consumer	United States	Henderson
3	CA-2017-138688	12/06/2017	16/06/2017	Second Class	DV-13045	Darrin Van Huff	Corporate	United States	Los Angeles
4	US-2016-108966	11/10/2016	18/10/2016	Standard Class	SO-20335	Sean O'Donnell	Consumer	United States	Fort Lauderdale
5	CA-2015-115812	09/06/2015	14/06/2015	Standard Class	BH-11710	Brosina Hoffman	Consumer	United States	Los Angeles
6	CA-2015-115812	09/06/2015	14/06/2015	Standard Class	BH-11710	Brosina Hoffman	Consumer	United States	Los Angeles
7	CA-2015-115812	09/06/2015	14/06/2015	Standard Class	BH-11710	Brosina Hoffman	Consumer	United States	West Jordan
8	CA-2015-115812	09/06/2015	14/06/2015	Standard Class	BH-11710	Brosina Hoffman	Consumer	United States	Los Angeles
9	CA-2015-115812	09/06/2015	14/06/2015	Standard Class	BH-11710	Brosina Hoffman	Consumer	United States	Los Angeles
10	CA-2015-115812	09/06/2015	15/05/2015	Standard Class	BH-11710	Brosina Hoffman	Consumer	United States	Los Angeles
11	CA-2015-115812	09/06/2015	14/06/2015	Standard Class	BH-11710	Brosina Hoffman	Consumer	United States	Los Angeles
12	CA-2015-115812	09/06/2015	14/06/2015	Standard Class	BH-11710	Brosina Hoffman	Consumer	United States	Los Angeles
13	CA-2015-115812	09/06/2015	14/06/2015	Standard Class	BH-11710	Andrew Allen	Consumer	United States	Concord
14	CA-2015-115812	09/05/2015	14/05/2015	Standard Class	BH-11710	Brosina Hoffman	Seond Class	United States	Fort Worth
15	CA-2015-115812	09/06/2015	14/06/2015	Standard Class	BH-11710	Brosina Hoffman	Seond Class	United States	Fort Worth
16	CA-2015-115812	09/06/2015	15/05/2015	Standard Class	BH-11710	Brosina Hoffman	Seond Class	United States	Los Angeles
17	CA-2015-115812	09/06/2015	14/06/2015	Standard Class	BH-11710	HP-14815	Consumer	United States	Los Angeles
18	CA-2015-199986	12/01/2015	15/01/2014	Standard Class	HP-14815	Harold Pawlan	Home Office	United States	Fort Worth
19	CA-2015-139896	12/01/2014	24/01/2014	Standard Class	HP-14815	Harold Pawlan	Home Office	United States	Fort Worth

Data Modeling

The dataset was used as a single table in Power BI. The model is simple and supports efficient analysis.



DAX Calculations

The following DAX measures were used:

Total Sales = SUM(Sales)
Average Sales = AVERAGE(Sales)
Order Count = COUNT(Order ID)
Unique Customers = DISTINCTCOUNT(Customer ID)
Max Sales = MAX(Sales) Min
Sales = MIN(Sales)

Total Sales	Average Sales	Order Count	Unique Customers	Max Sa	Sa
$2,297,201	$229.86	9,994	793	$22,638	$0.44

Dashboard Development

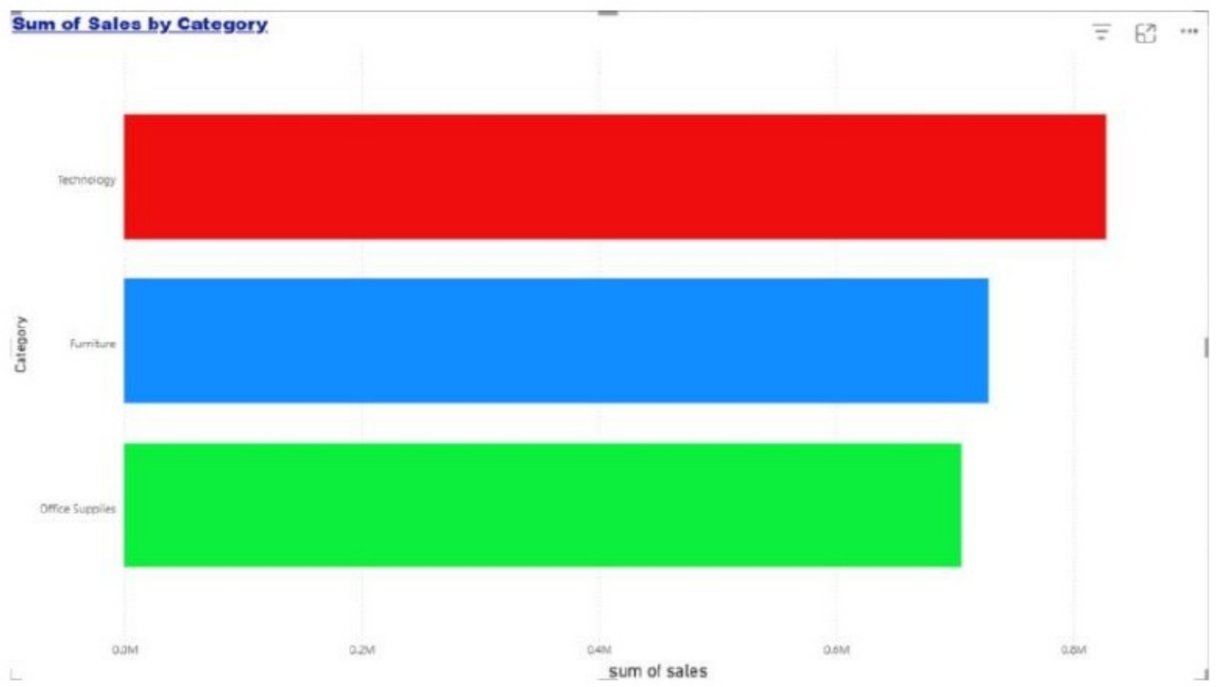
The dashboards include:

Overview Dashboard (Card, Bar Chart, Pie Chart)
Trend Analysis (Line Chart)
Category Analysis (Bar Chart)

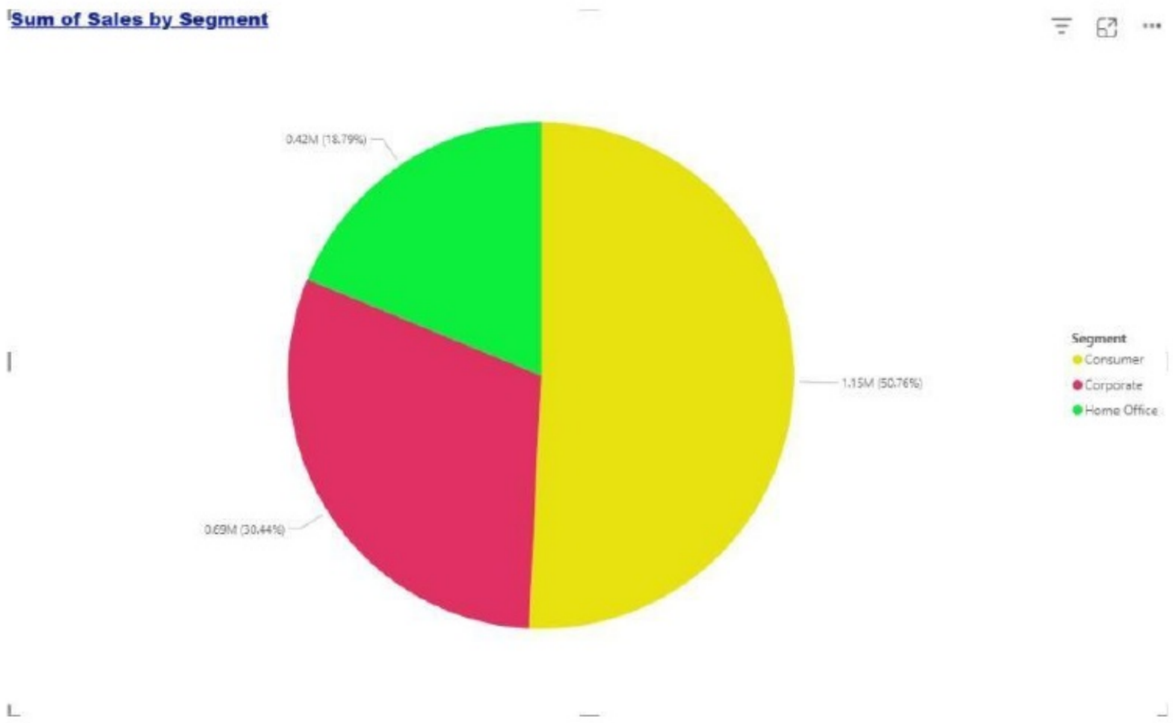
Geographical Report (Map) Detailed
Report (Table)

Slicers were used to provide interactivity and filtering.

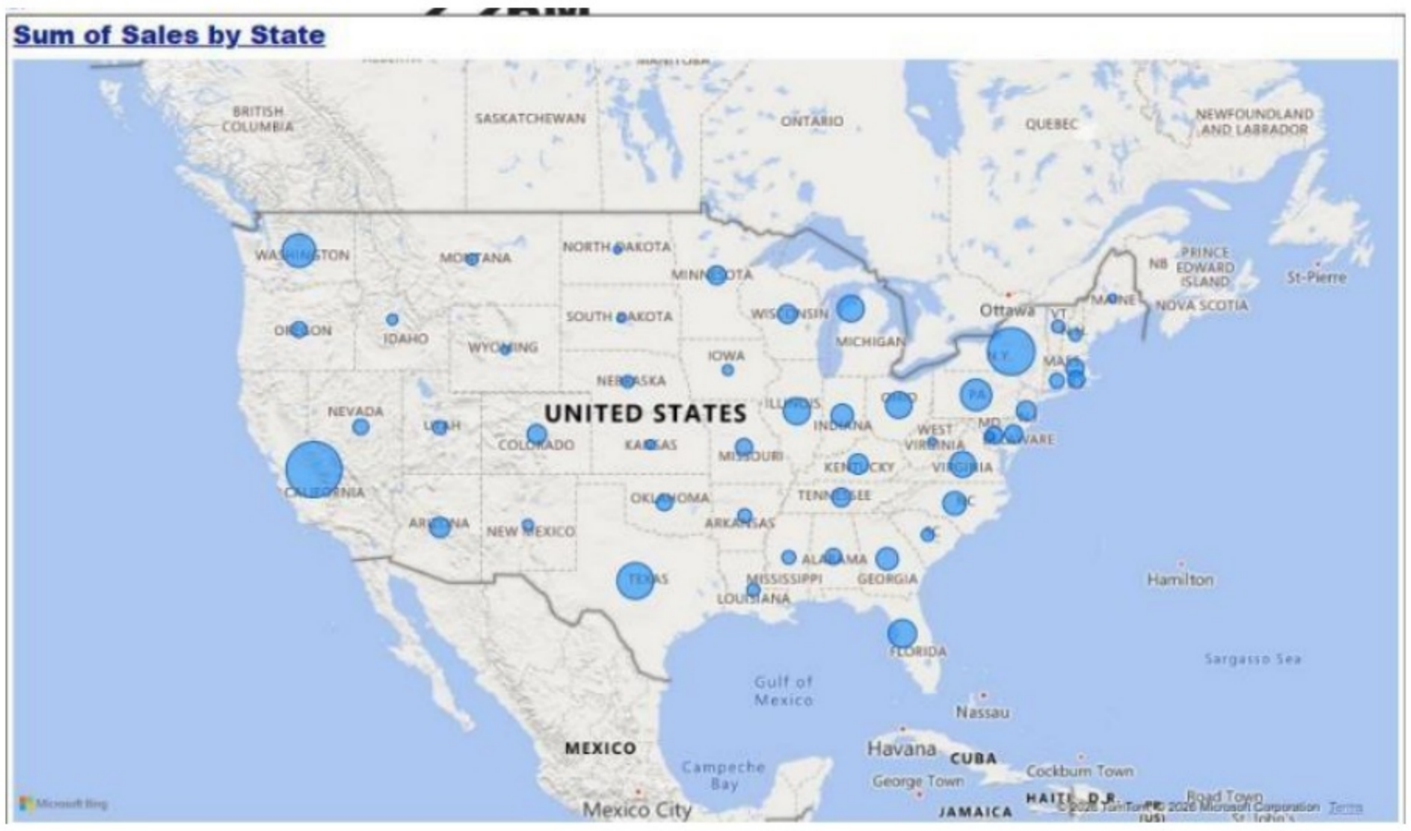
Sales by Category:



Sales by Segment:



Sales by Region:



Sales Trend Over Time:



Detailed Report:

Order ID	Customer Name	City	Category	Segment	Sum of Sales
CA-2015-100006	Dennis Kane	New York City	Technology	Consumer	377.97
CA-2015-100090	Ed Braxton	San Francisco	Furniture	Corporate	502.49
CA-2015-100090	Ed Braxton	San Francisco	Office Supplies	Corporate	196.70
CA-2015-100293	Neil Französisch	Jacksonville	Office Supplies	Home Office	91.06
CA-2015-100328	Jasper Cacioppo	New York City	Office Supplies	Consumer	3.93
CA-2015-100363	Jim Mitchum	Glendale	Office Supplies	Corporate	21.38
CA-2015-100391	Barry Weirich	New York City	Office Supplies	Consumer	14.62
CA-2015-100678	Kunst Miller	Houston	Furniture	Consumer	317.06
CA-2015-100678	Kunst Miller	Houston	Office Supplies	Consumer	152.04
CA-2015-100678	Kunst Miller	Houston	Technology	Consumer	227.98
CA-2015-100706	Laurel Elliston	Springfield	Furniture	Consumer	29.46
CA-2015-100706	Laurel Elliston	Springfield	Technology	Consumer	99.98
CA-2015-100762	Nat Gilpin	Jackson	Office Supplies	Corporate	508.62
CA-2015-100860	Cindy Stewart	Pomona	Office Supplies	Consumer	18.75
CA-2015-100867	Eugene Hildebrand	Lakewood	Technology	Home Office	321.55
CA-2015-100881	Daniel Raglin	Albuquerque	Technology	Home Office	302.38
CA-2015-100895	Stewart Visinsky	Roswell	Office Supplies	Consumer	365.50
CA-2015-100895	Stewart Visinsky	Roswell	Technology	Consumer	239.97
CA-2015-100916	Frank Hawley	Newport News	Furniture	Corporate	591.32
CA-2015-100916	Frank Hawley	Newport News	Office Supplies	Corporate	197.54
CA-2015-100972	Dennis Bolton	Salt Lake City	Office Supplies	Home Office	166.44
CA-2015-101147	Matt Collins	Chicago	Office Supplies	Consumer	2.39
Total					**2,261,536.78**

Challenges Faced

Difficulty in selecting dataset
Errors in DAX formulas
Map visualization initially disabled
Understanding Power BI interface

Conclusion

This project demonstrates how Power BI can be used to analyze and visualize data. The dashboards provide clear insights into sales performance and help in better decision-making.

Future Enhancements

Integration with real-time data
Advanced analytics using AI
More complex data modeling

References

https://www.kaggle.com
Microsoft Power BI Documentation